

WASP-43b

R-0346

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260204 · created 2026-07-18T08:25:44+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461075.9492 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.023
Transit depth (Rp/Rs)^2	3.12 +/- 0.83 [%]
Orbital Inclination (inc)	80.82 +/- 1.99
Airmass coefficient 1 (a1)	0.959 +/- 0.016
Airmass coefficient 2 (a2)	0.0254 +/- 0.0087
Scatter in the residuals of the lightcurve fit is	1.71 %
Best Comparison Star	#3 - [463, 301]
Optimal Method	PSF photometry
Transit Duration (day)	0.0463 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.177 ± 0.023 vs 0.1594 (deviation 0.77σ)
Duration fit vs published	0.0463 d vs 0.0512 d (deviation 0.52σ)
Mid-time O-C	1.2 min
Depth SNR	7.7
Residual scatter	1.71 %

Light curve

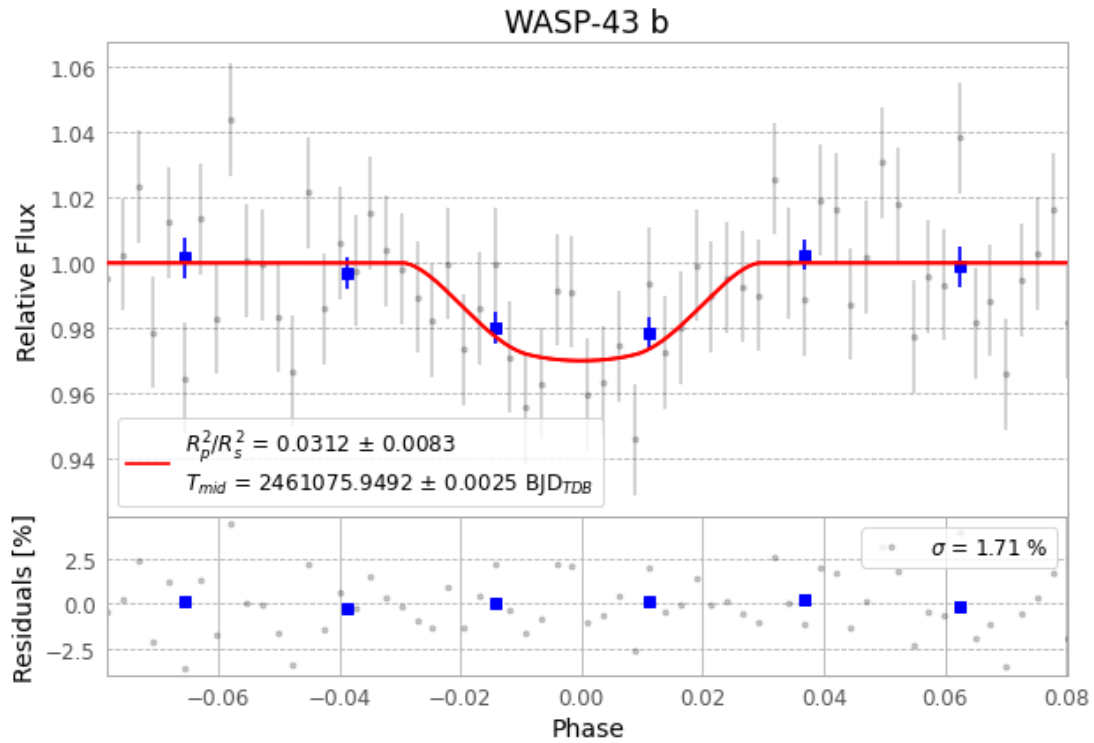


Figure 1 — FinalLightCurve_WASP-43 b_2026-02-04.png (SHA-256 927719cb53015436...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [441, 173], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e4bfe8e8be065869...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

63 FITS frames (41 MB), WASP-43260204090616.FITS → WASP-43260204121216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260204.log (62a54d43e819...), FinalLightCurve_WASP-43 b_2026-02-04.png (927719cb5301...), FinalLightCurve_WASP-43 b_2026-02-04.csv (b7b77035b61b...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 08:25Z.